

Understanding How Interfaces Affect Users

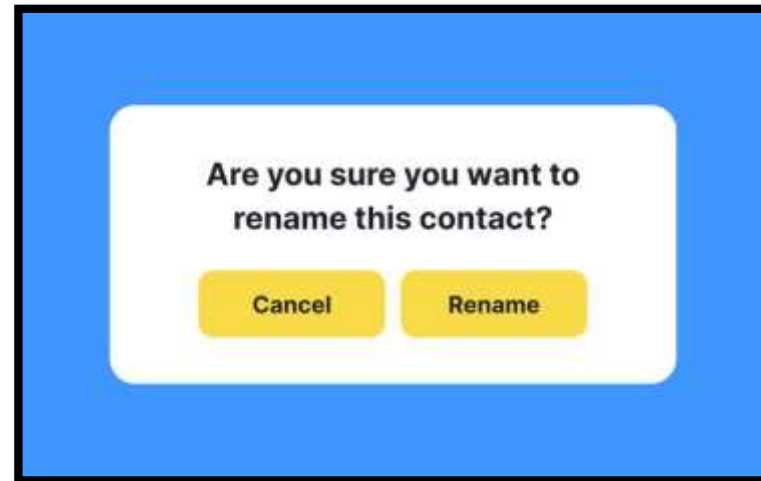
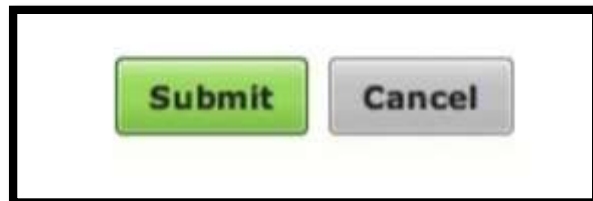


Overview

- Expressive interfaces
- Negative aspects
- Anthropomorphism and interface agents



Can a simple button design affect how fast you complete a task?



Expressive Interfaces

- Expressive forms like emoticons, icons, sounds, and virtual agents .
- The style of an interface, in terms of the shapes, fonts, colors, balance, and white space can influence emotional impact.

Expressive Interfaces

- How to design interactive systems to make people **respond** in certain ways
 - Like be happy, to be trusting, to learn, to be motivated



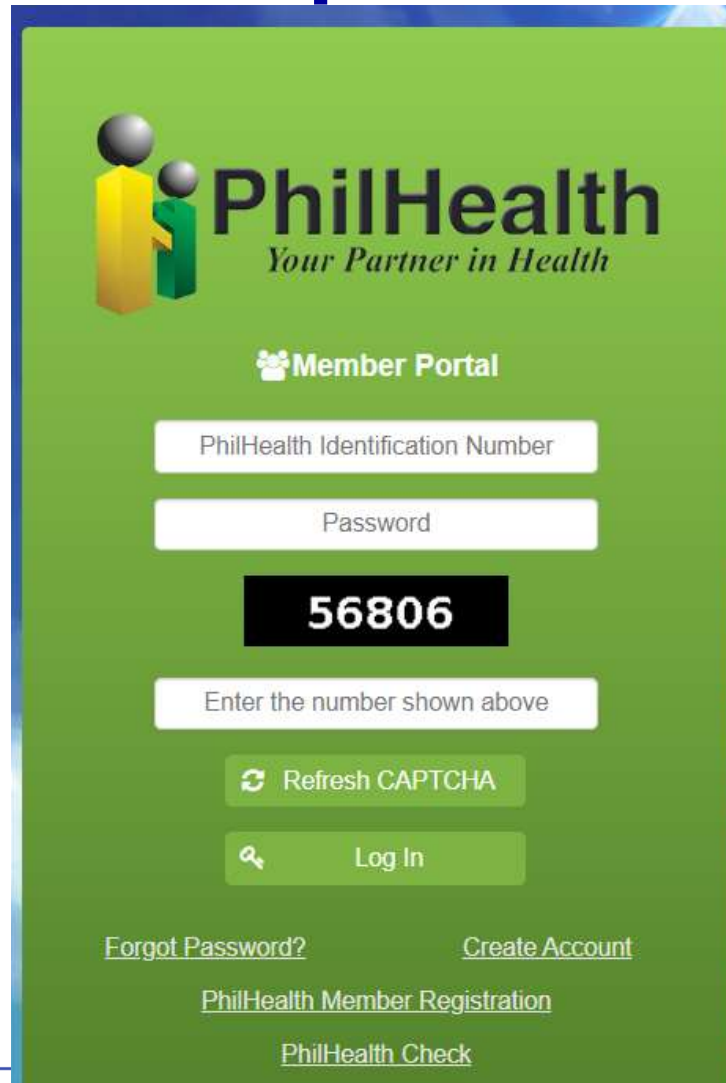


Expressive Interfaces

- Color, icons, sounds, graphical elements and animations are used to make the **'look and feel'** of an interface appealing



Can a simple button design affect how fast you complete a task?



The image shows a screenshot of the PhilHealth Member Portal login interface. The background is a solid green color. At the top left, there is a logo consisting of two stylized human figures, one yellow and one green, with black spheres for heads. To the right of the logo, the text "PhilHealth" is written in a large, bold, black font, and below it, "Your Partner in Health" is written in a smaller, italicized black font. Below the logo and text, the words "Member Portal" are displayed in a white font, preceded by a small white icon of a person. The login form consists of several white input fields and buttons. The first field is labeled "PhilHealth Identification Number". The second field is labeled "Password". Below the password field, there is a black rectangular box containing the white text "56806". Below this box is a white input field labeled "Enter the number shown above". Below this field is a green button with a white circular arrow icon and the text "Refresh CAPTCHA". Below that is a green button with a white magnifying glass icon and the text "Log In". At the bottom of the form, there are three links: "Forgot Password?", "Create Account", and "PhilHealth Member Registration". At the very bottom, there is a link "PhilHealth Check".

PhilHealth
Your Partner in Health

Member Portal

PhilHealth Identification Number

Password

56806

Enter the number shown above

Refresh CAPTCHA

Log In

[Forgot Password?](#) [Create Account](#)

[PhilHealth Member Registration](#)

[PhilHealth Check](#)



Can a simple button design affect how fast you complete a task?

The screenshot displays the Craigslist Philippines interface for Davao City. The layout includes a top navigation bar with the site logo, location, and user actions. The main content area is organized into several columns of category buttons. On the left, there is a sidebar with a search bar, an event calendar, and links to help and about pages. On the right, a vertical menu lists regional and global categories.

Top Bar: craigslist^{ph} | davao city | faves | post | acct | english ▾

Left Sidebar:

- post an ad
- search craigslist
- event calendar**
S M T W T F S
5 6 7 8 9 10 11
12 13 14 15 16 17 18
19 20 21 22 23 24 25
26 27 28 29 30 1 2
- help, faq, abuse, legal
avoid scams & fraud
personal safety tips
- about craigslist
best-of-craigslist
what's new
system status

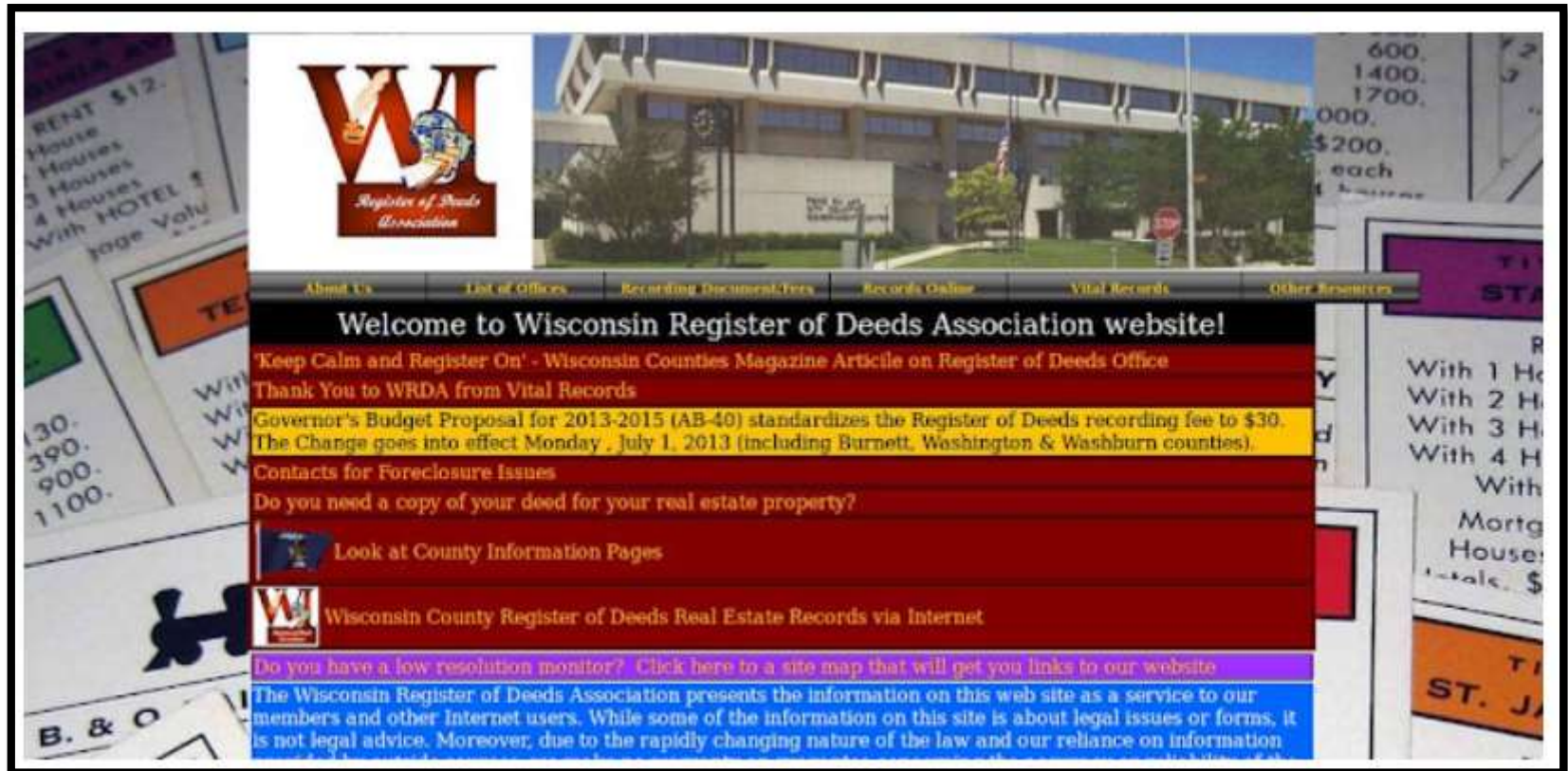
Main Content Area:

- community**
activities | lost+found
artists | missed
childcare | connections
classes | musicians
events | pets
general | politics
groups | rants & raves
local news | rideshare
volunteers
- housing**
apts / housing
housing swap
housing wanted
office / commercial
parking / storage
real estate for sale
rooms / shared
rooms wanted
sublets / temporary
vacation rentals
- for sale**
antiques | farm+garden
appliances | free
arts+crafts | furniture
atv/utv/sno | garage sale
auto parts | general
aviation | heavy equip
baby+kid | household
barter | jewelry
beauty+hlth | materials
bike parts | motorcycle parts
bikes | motorcycles
boat parts | music instr
boats | photo+video
books | rvs+camp
business | sporting
- services**
automotive | labor/move
beauty | legal
cell/mobile | lessons
computer | marine
creative | pet
cycle | real estate
event | skilled trade
farm+garden | sm biz ads
financial | travel/vac
health/well | write/ed/tran
household
- jobs**
accounting+finance
admin / office
arch / engineering
art / media / design
biotech / science
business / mgmt
customer service
education
etc / misc
food / bev / hosp
general labor
government
human resources
legal / paralegal
manufacturing
marketing / pr / ad
medical / health
nonprofit sector
real estate
retail / wholesale
sales / biz dev
salon / spa / fitness
security
skilled trade / craft
software / qa / dba
systems / network
technical support

Right Sidebar:

- philippines
bacolod
bicol region
cagayan de oro
cebu
davao city
ililo
manila
pampanga
zamboanga
- asia/pacific
- ci worldwide

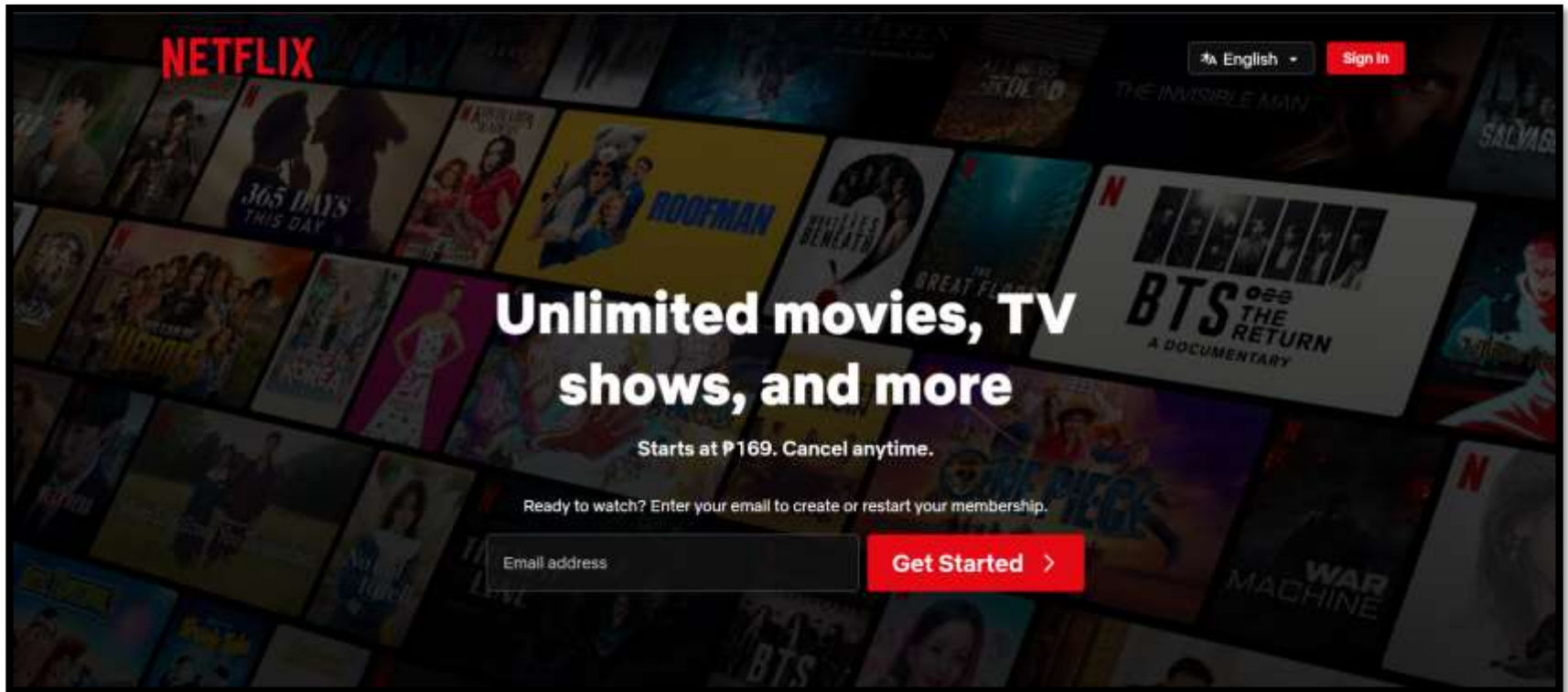
Can a simple color design affect how fast you complete a task?



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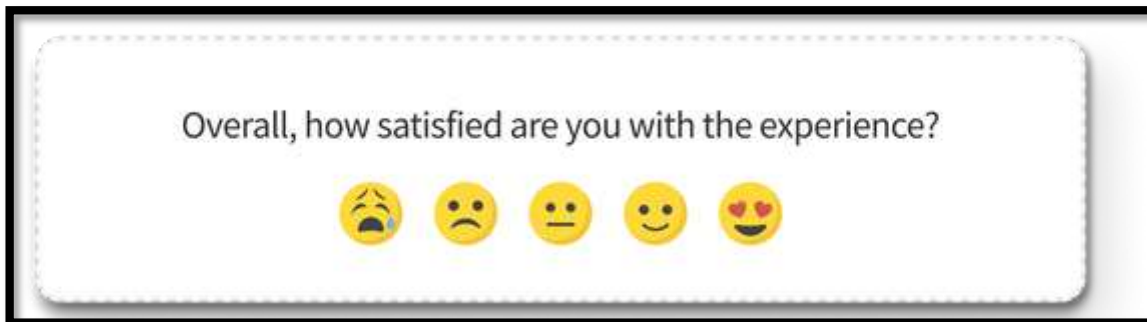
Friendly Interfaces

- Make users feel more at ease and comfortable



User-created Expressiveness

- Users have created *emoticons* - compensate for lack of expressiveness in text communication:



Possible causes of User frustration when interacting with an application or system?

User frustration

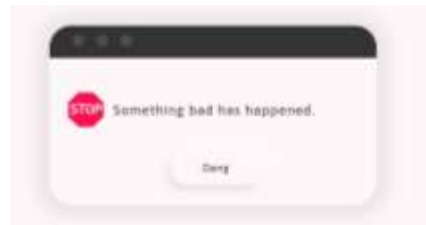
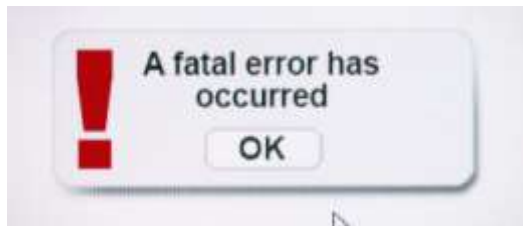
Many causes:

- When an application doesn't work properly or crashes
- When a system doesn't do what the user wants it to do
- When a user's expectations are not met
- When a system does not provide sufficient information to enable the user to know what to do
- When error messages pop up that are vague, obtuse or condemning
- When the appearance of an interface is garish, noisy, gimmicky or patronizing

Error messages

Shneiderman's guidelines for error messages include:

- avoid using terms like FATAL, INVALID, BAD
- Audio warnings
- Avoid UPPERCASE and long code numbers
- Messages should be precise rather than vague
- Provide context-sensitive help

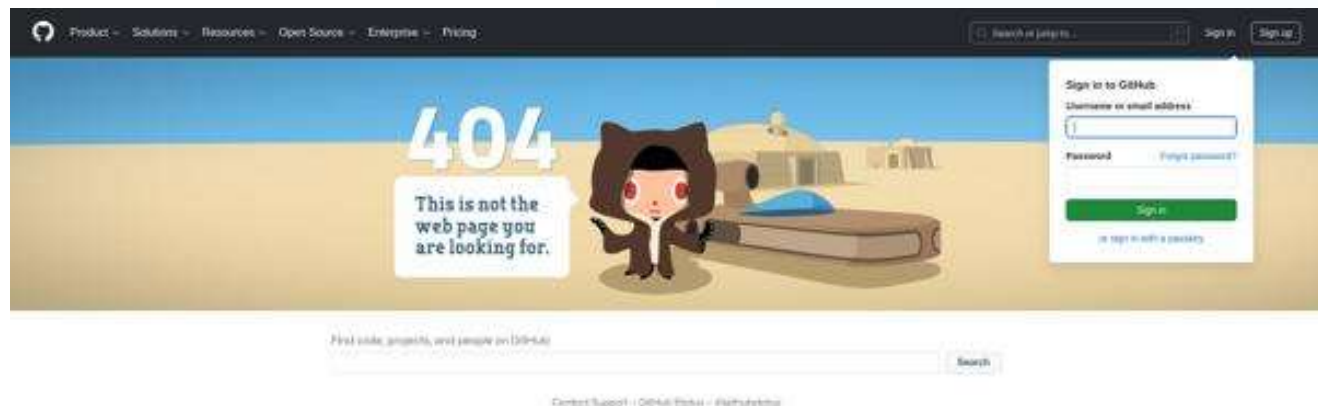
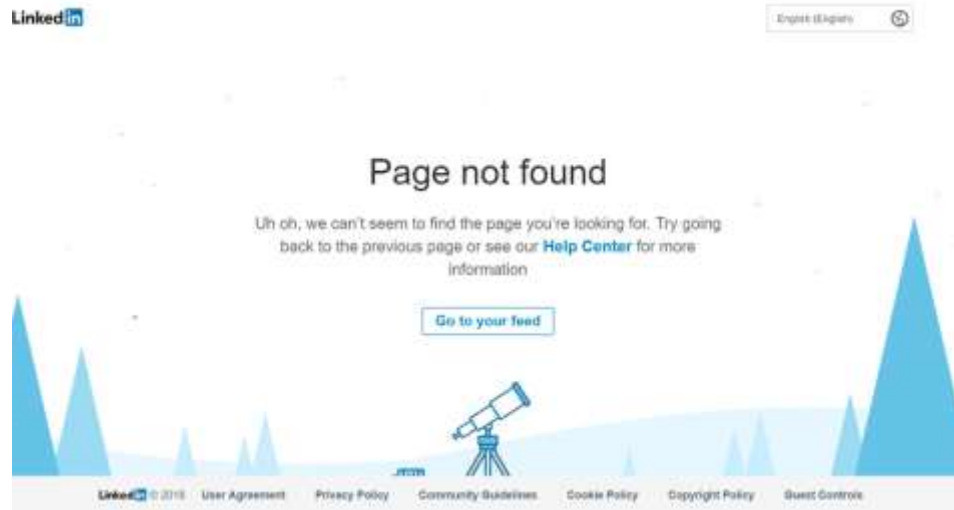


Error text clearly communicates the mistake and describes how to correct it



Lack of information may lead to misunderstanding

Website Error message...

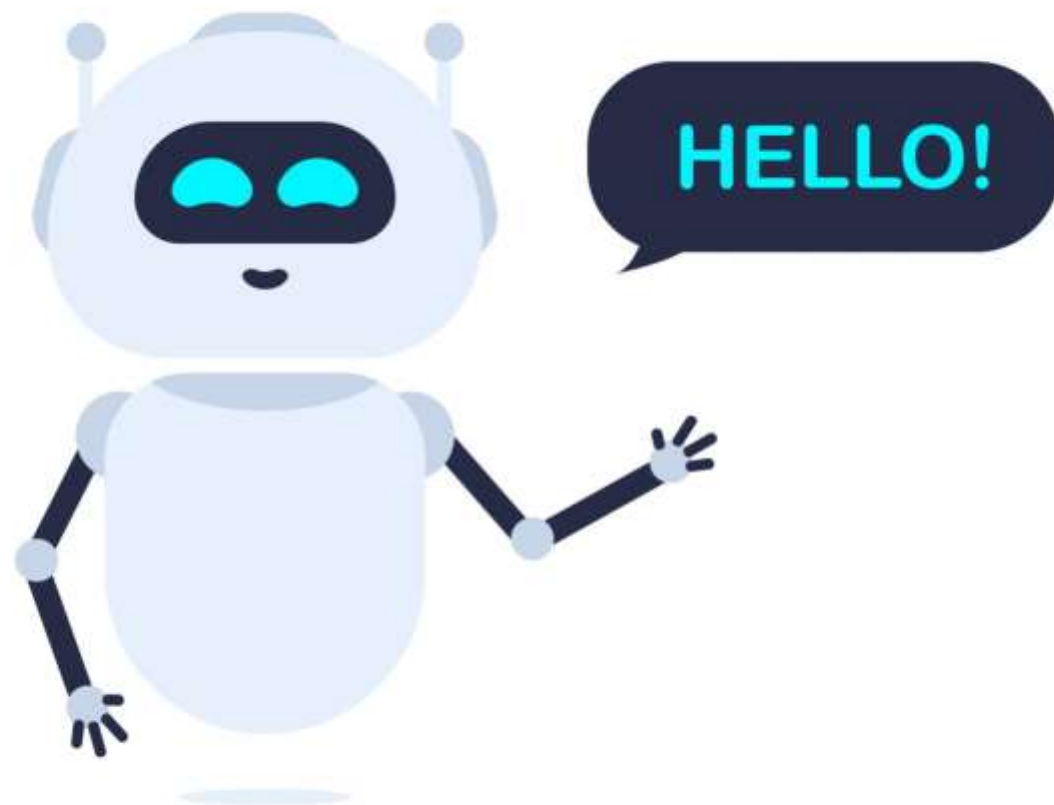


Should Computers say they're sorry?

Should Computers say they're sorry?

- Reeves and Naas (1996) argue that computers should be made to apologize.
- Should emulate human etiquette
- How sincere would they think the computer was being? For example, after a system crash:
 - “I’m really sorry I crashed. I’ll try not to do it again”

A study comparing error messages that apologize versus those that did not were perceived by users less frustrating(Part, 2012)

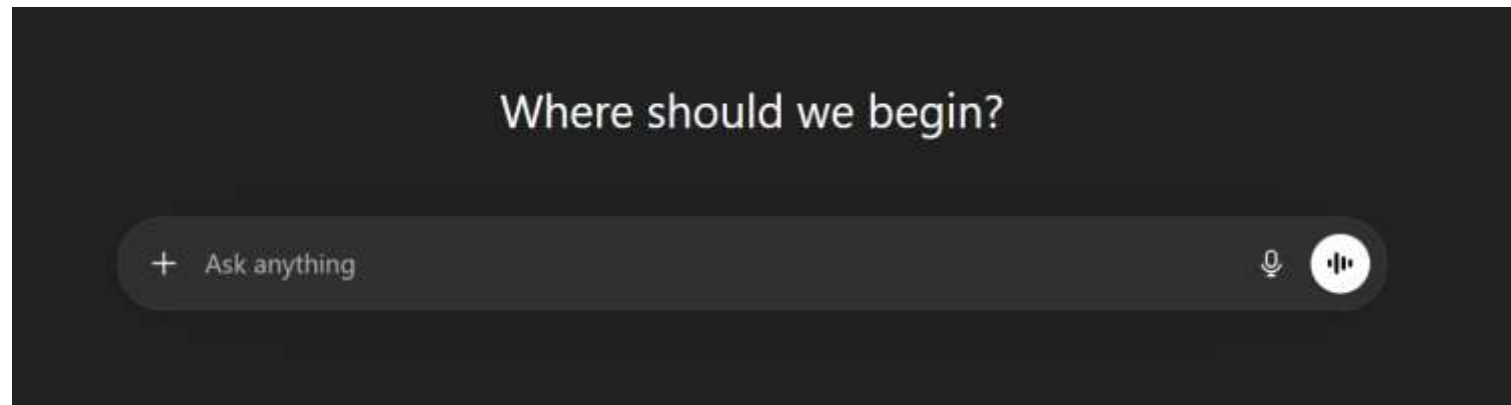


Anthropomorphism

- Attributing human-like qualities to inanimate objects.
- Making technology feel human-like to improve user experience.
 - Helps users understand interactions faster
 - Creates emotional connection with users

Anthropomorphism

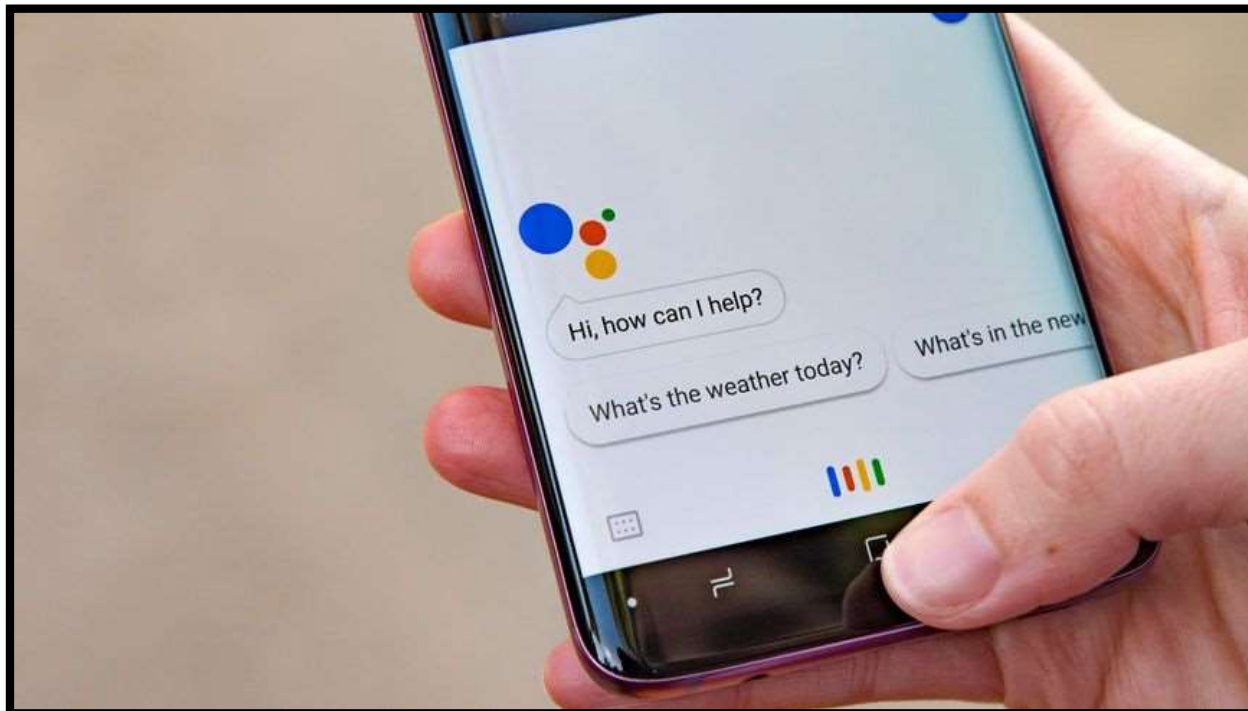
- Example: **A chatbot that says:**



- A human-like way of asking what the user wants to do
- Politely inviting the user to start the interaction
- Giving a sense of guidance and friendliness

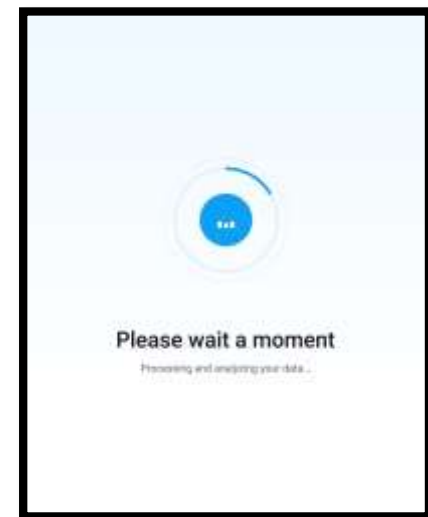
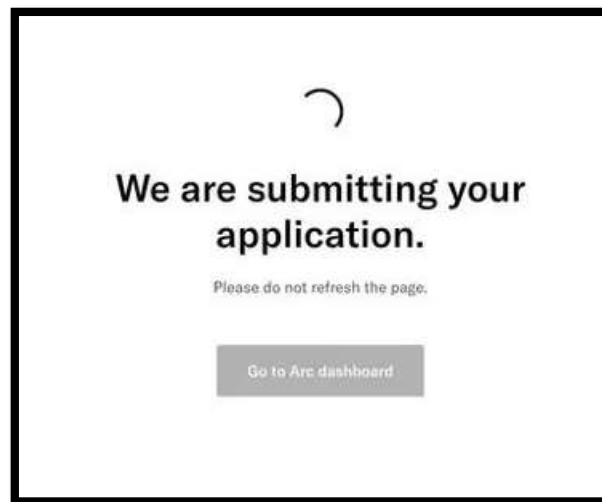
Anthropomorphism

- Example: **Voice assistants like Siri or Google Assistant** speaking in a natural human voice



Anthropomorphism

- A **loading screen** with a character saying: *"I'm working on it!"*
- **Error messages** like: *"Oops! Something went wrong."*



**What are the types of interfaces used
when giving feedback after a user
gets something wrong?**

Which do you prefer?

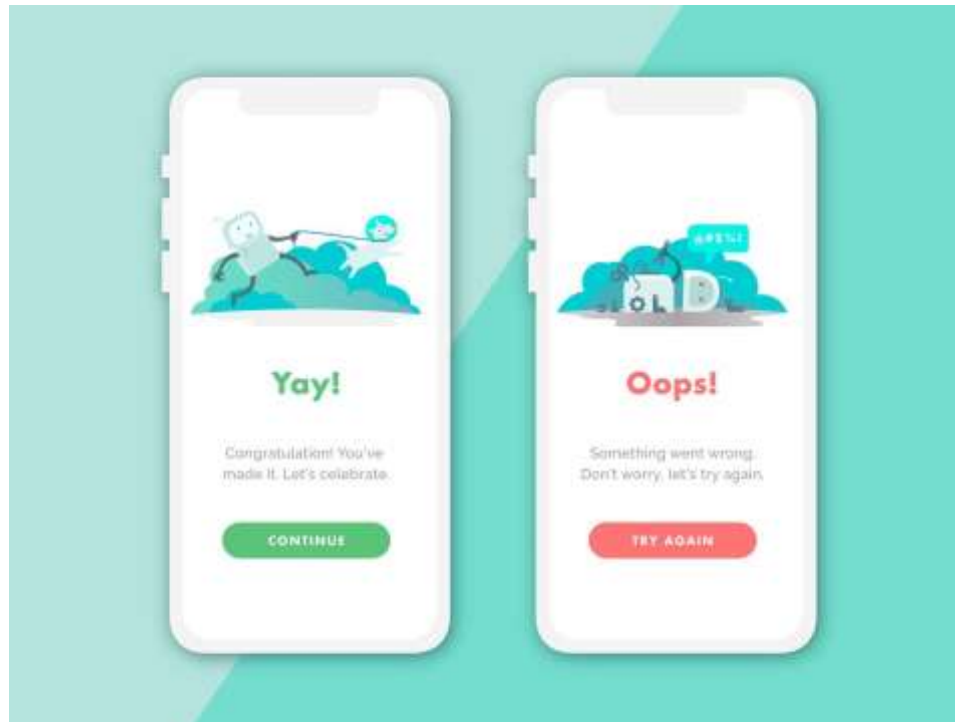
2. Feedback when get something wrong

1. "Now Carla, that's not right. You can do better than that. Try again!"
1. "Incorrect. Try again!"

Is there a difference as to what you prefer depending on type of message? Why?

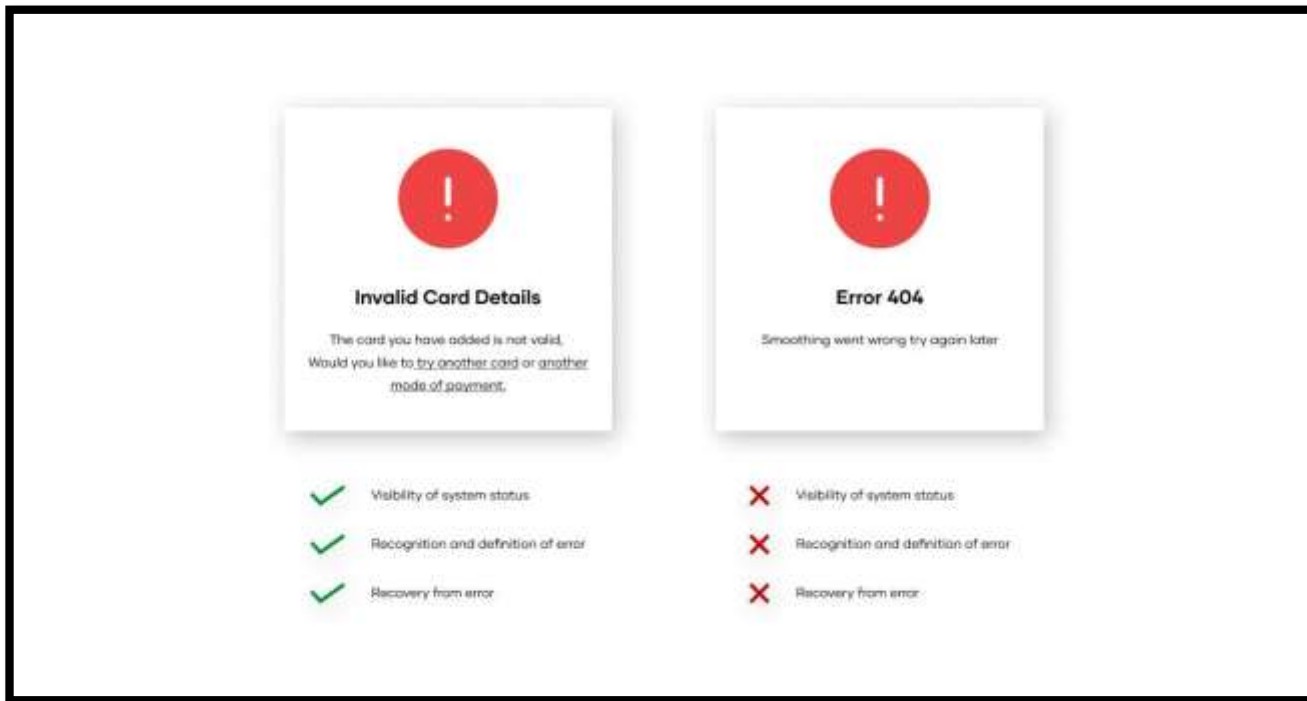
Which do you prefer?

2. Feedback when get something wrong



Which do you prefer?

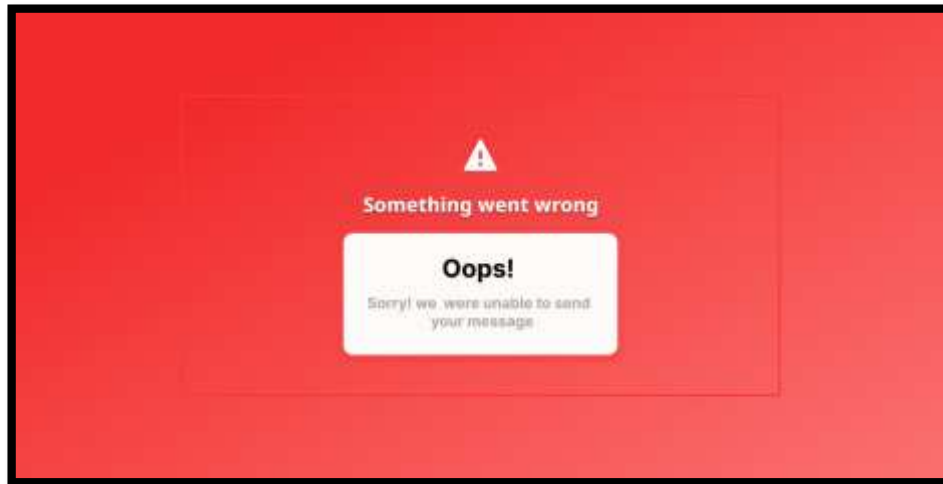
2. Feedback when get something wrong



Which do you prefer?

2. Feedback when get something wrong

?

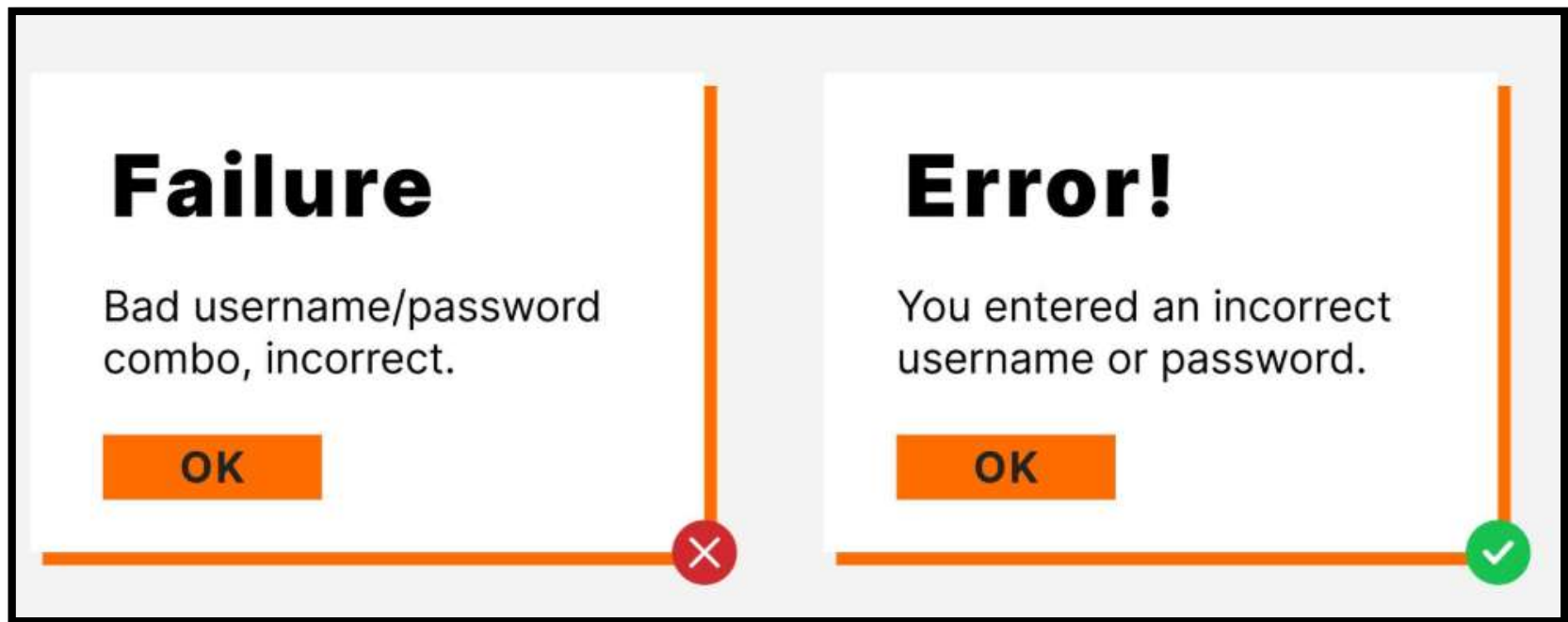


?



Which do you prefer?

2. Feedback when get something wrong



The Power of Positive Feedback

Reeves & Nass (1996) found that computers that flatter and praise users in educational software have a **positive impact**.

Example Feedback

“Your question makes an important and useful distinction. Great job!”



Impact on Students



More Motivated



Willing to Continue Exercises



Engaged Learning



Friendly Praise Helps Students Succeed!

Criticism of Anthropomorphism

Using overly human-like systems can be problematic.

Risks of Anthropomorphism



Deceptive



Anxious & Inferior



Feel Stupid



"Now Chris, that's not right. You can do better than that. Try again."

Studies Show



Personalized feedback is considered:

- Less Honest
- Reduced Responsibility

Incorrect. Try again.



More Impersonal



Less Honest

(Quintanar, 1982)



Reduced Responsibility



Many prefer neutral, factual responses

Criticism of Anthropomorphism

Studies have shown that personalized feedback is considered to be less honest and makes users feel less responsible for their actions (e.g. Quintanar, 1982).

- Anthropomorphism is not always good.
- When systems sound too personal or emotional, users may feel uncomfortable or judged.
- **Better to keep feedback clear, neutral, and direct so users feel.**



How bad error messages are harming your users

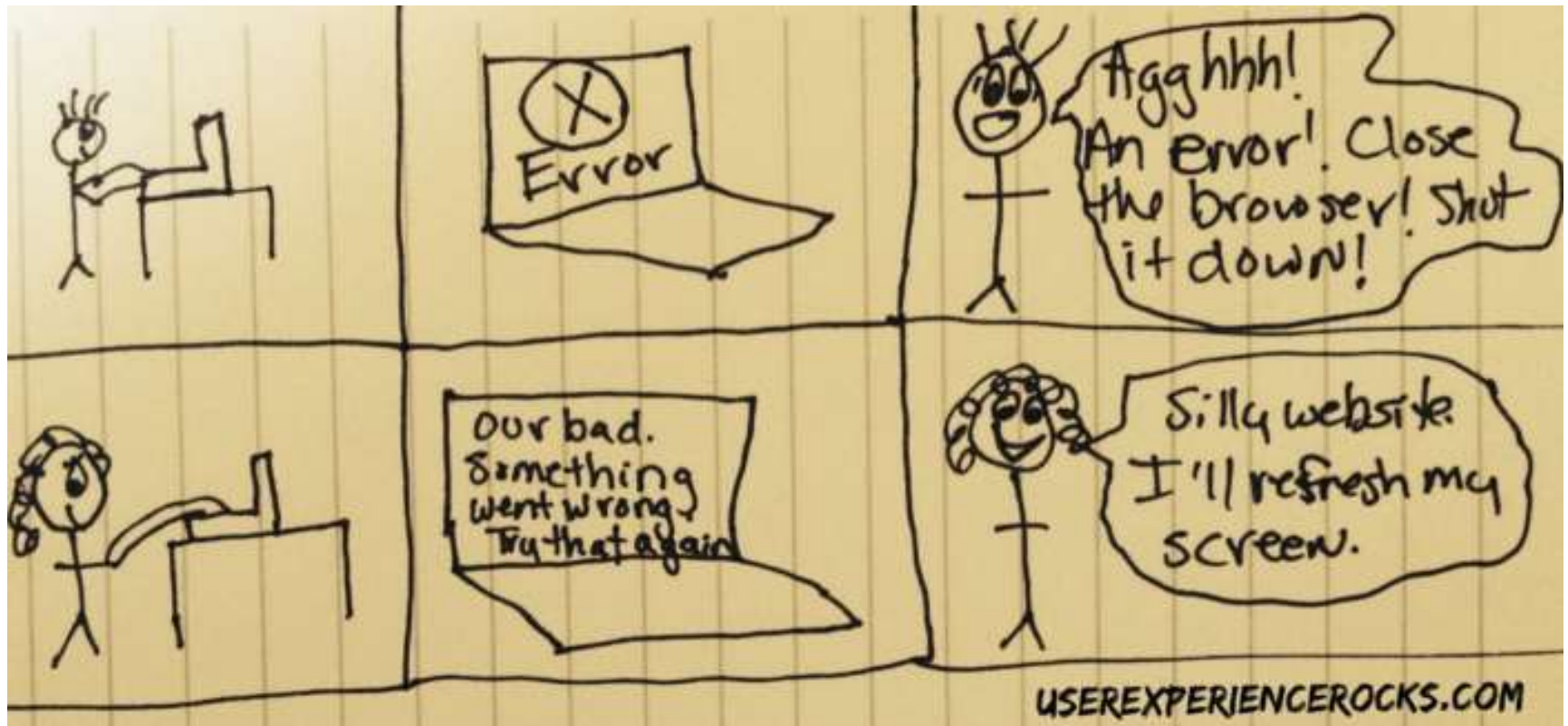
Author: Kayla J Heffernan

<https://uxdesign.cc/bad-error-messages-are-harming-your-users-1ee52b8e520f>

Do's And Don'ts For Better Error Messages.

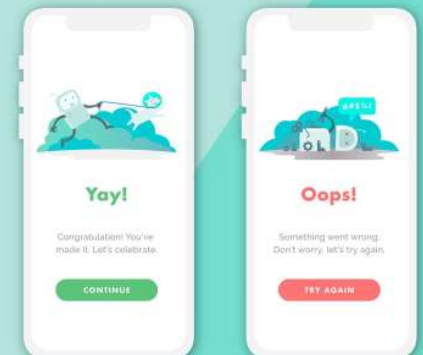
- Research from 2009 (Riedl, R et al, 2009) showed error messages can increase cortisol (the stress hormone) levels in individuals when they were interacting with computers that faced a system breakdown.
- Our error messages are increasing stress and anxiety.

Illustration by Jennifer Aldrich of User Experience Rocks



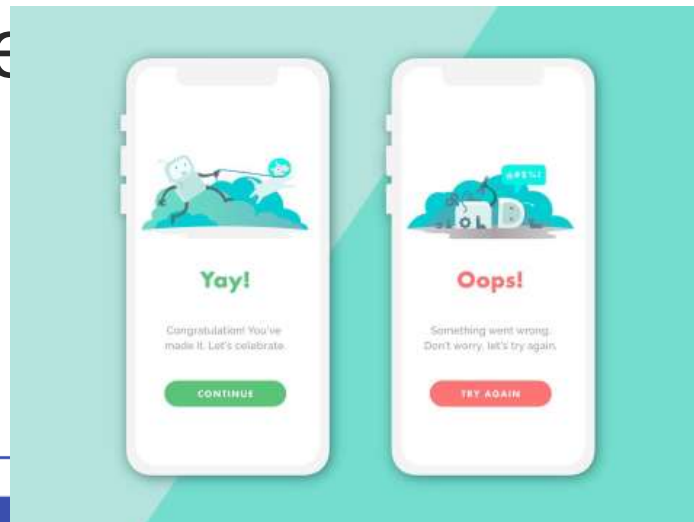
The Importance of Good Error Messages

- We have a responsibility to carefully consider our error messages as to not harm our users.
- Good error messages are particularly important as they:
 - A. Increase speed of use;
 - B. Increase users subjective satisfaction;
 - C. Reduce repeat errors and complaints while increasing the learnability of the system.



What does the Good Error advantages mean?

- Users can 'self-serve' and correct errors allowing them to complete their goal.
- This also saves on support costs if fewer people are raising tickets or calling a helpdesk.



How do you write good error messages?

- Errors happen. But the golden rule should be prevention first.
- Make it easy for your users to input the data in the correct format and explain to them exactly what they need to do.
- Errors should be a last resort.

Do the following

- Be explicit and indicate something has gone wrong
- Include messages on same screen as where the error occurred
- Include the product or component name. This helps

Do.

PERSONAL INFO

First name
James

Last name
Connors

Password*

• Must be at least 8 characters long
• Use at least one digit (0-9) and special characters like (#, @, \$ %)
• Use lowercase and uppercase letters (a-z)

Don't.

PERSONAL INFO

First name
James

Last name
Connors

Password*

Not valid

Do the following

- Make error messages as short as possible while communicating all needed information.
- Explain how to fix the error
- Use the user's language
- Use the correct combination of buttons.

Do.

PERSONAL INFO

First name
James

Last name
Connors

Password*

• Must be at least 8 characters long
• Use at least one digit (0-9) and special characters like (#, @, \$ %)
• Use lowercase and uppercase letters (a-z)

Don't.

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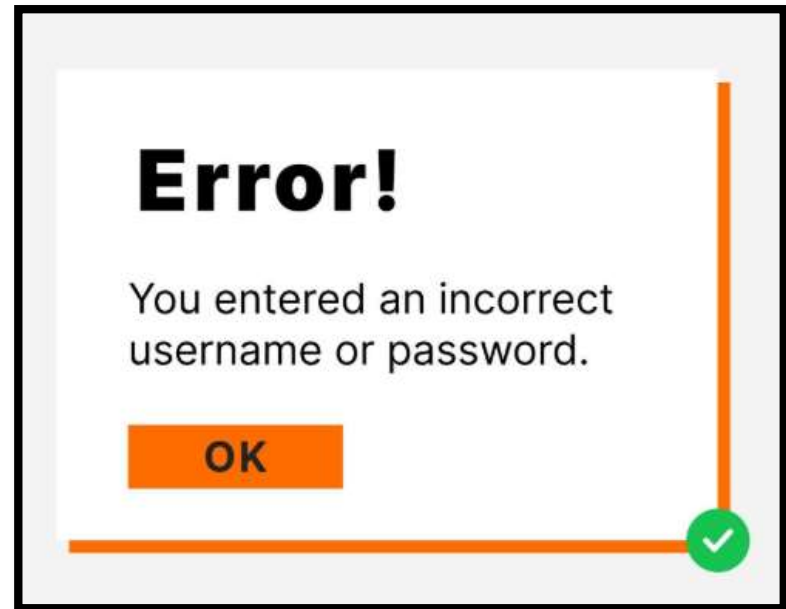
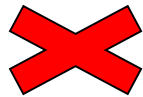
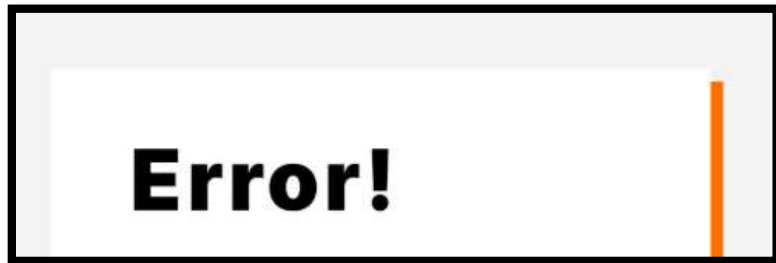
Not valid

Don't do the following

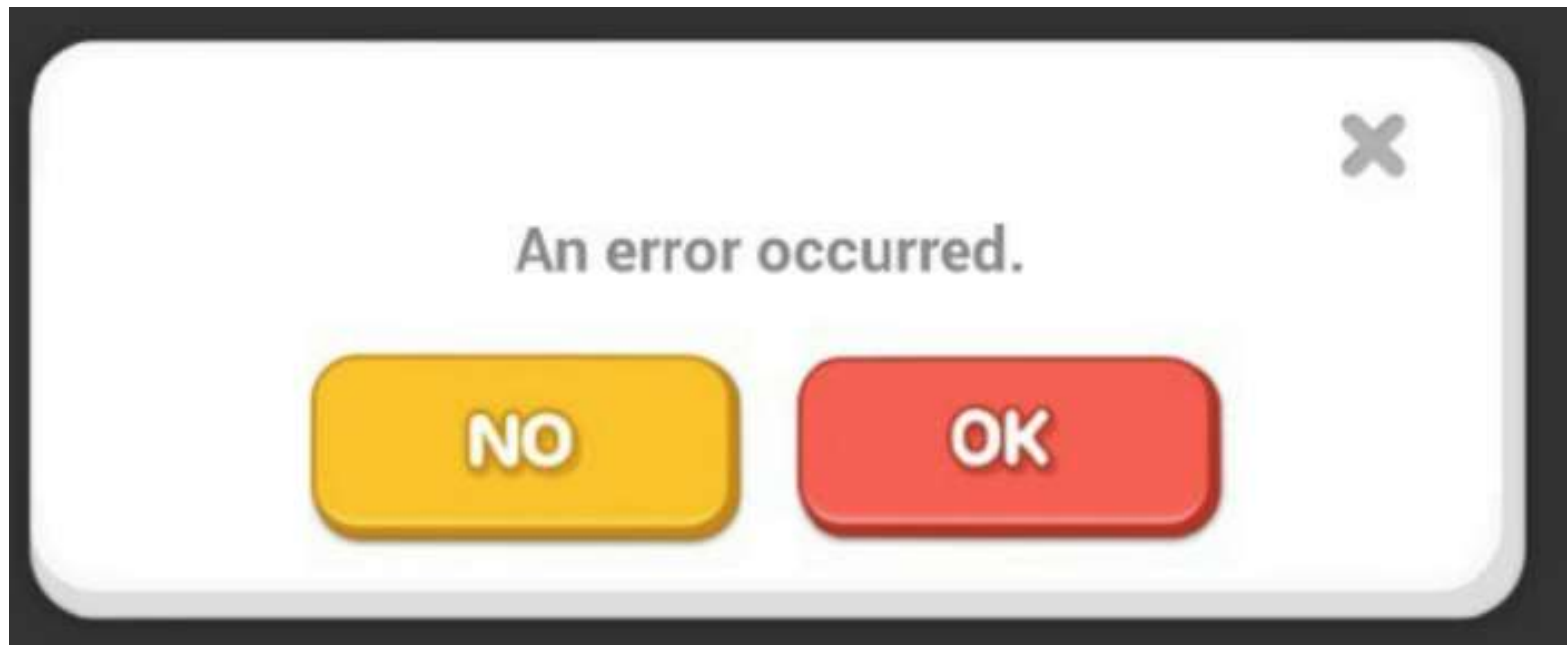
- Write like 'a programmer' — avoid jargon.
- Imply the error is user's fault (e.g. "illegal command").
- Over-communicate the problem
- Give unnecessary error messages

Don't do the following

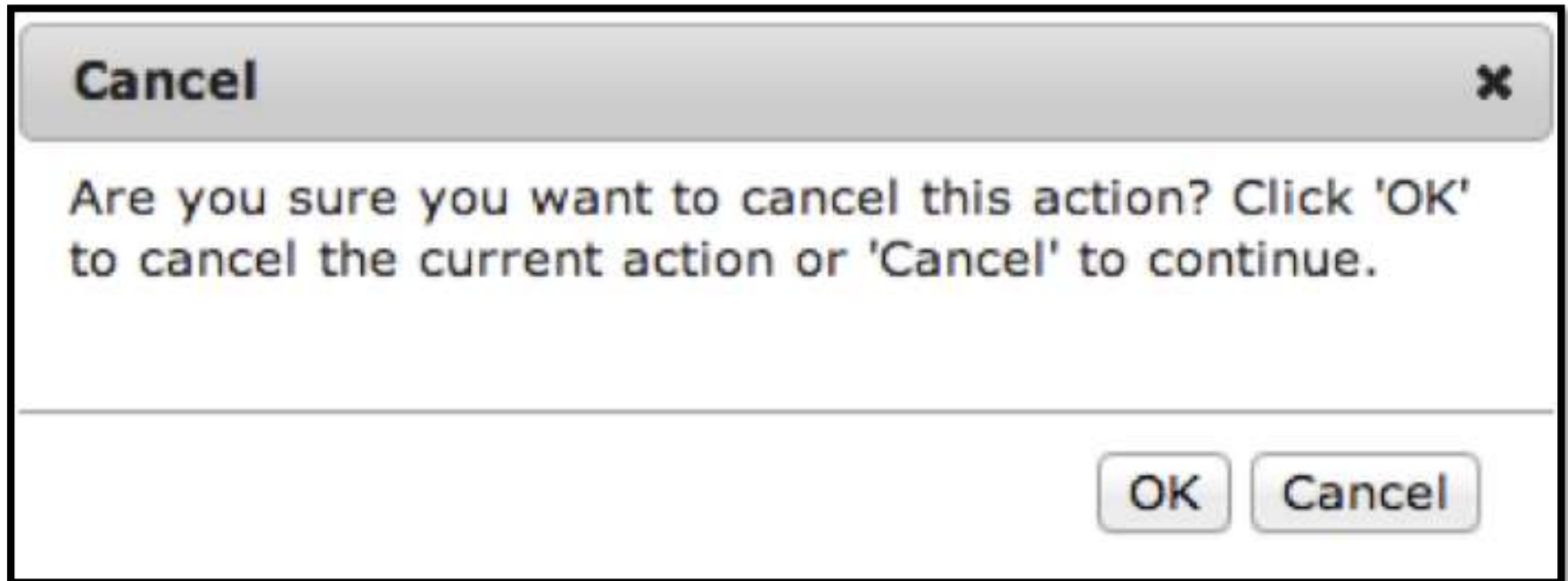
- Include the word “error” in the title bar if you can be more specific



Examples of Bad Error Messages



Examples of Bad Error Messages



Examples of Bad Error Messages



Readings about error messages

- <https://cxl.com/blog/error-messages/>
- <https://uxdesign.cc/bad-error-messages-are-harming-your-users-1ee52b8e520f>

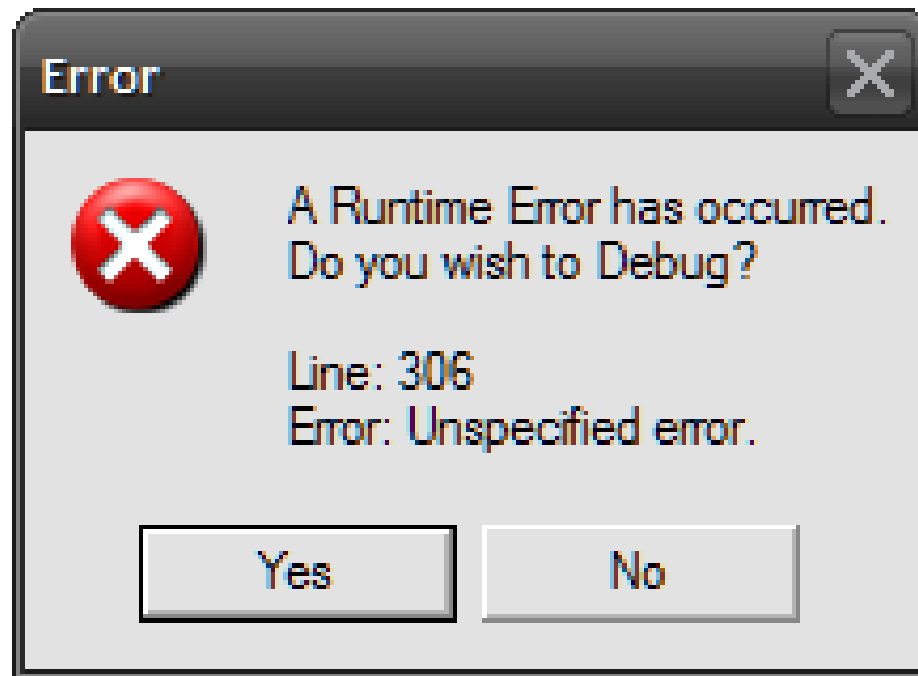
Activity

1. Explain what's wrong to the following error messages.



Activity

2. Explain what's wrong to the following error messages.



Activity

3. Explain what's wrong to the following error messages.

